europass



# Luiz Felipe Couto Gontijo

**Nationality:** Brazilian **Date of birth:** 16/09/1999 **Gender:** Male **Phone:** (+55) 31988444681 (Mobile) **Email address:** luizcouto@dcc.ufmg.br **Whatsapp Messenger:** 5531988444681 **LinkedIn:** www.linkedin.com/in/luiz-felipe-couto **Github:** https://github.com/luiz-couto **Address:** Av. Veredas das Gerais, 631, Condominio Veredas das Gerais, 34012000, Nova Lima, Brazil (Home)

## ABOUT MYSELF

Computer Science graduate from the Federal University of Minas Gerais (UFMG), currently working as a Senior Software Engineer and doing a Master's degree in Games Engineering at University of Warwick, I'm passionate about software development, always looking to learn new technologies, meet new people and new areas of innovation. I have experience as a Full Stack developer of Systems for Web and Mobile applications, including HTML5, CSS, Javascript, Typescript, Node js, React / React Native and Flutter. I also worked with the development of projects using Go and related to microsservices, involving maintenance and development of backend API, networking and messaging services, using technologies such as Kafka, Redis, Websockets, Kubernetes, GCloud, Docker and so on. As a Computer Science student, I have experience in Software Engineering with C, C++, Data Science (Python) and in Compilers (LLVM). My final paper in college was a project in Rust and C++, in which I developed a series of parallel, distributed and efficient image processing algorithms.

## EDUCATION & TRAINING

### Games Engineering Master's (In Progress)

***University Of Warwick*** [ 27/09/2025 - Current ]

**City:** Coventry | **Country:** United Kingdom

**Field(s) of study:** Computer Graphics: ; Games Engineering: ; Computer Science: | **Level in EQF:** 7

As a Games Engineering master's student at the University of Warwick, I developed the skills to understand and implement a graphics engine from the ground up. This includes algorithms such as Rasterization, Rendering, Ray tracing, and Path Tracing. My projects covered 2D and 3D techniques using C++ and DirectX. Game development was also a focus, with individual and group projects where we built games using our own engine, as well as Unreal Engine and Godot.

### Computer Science Bachelor's

***Universidade Federal de Minas Gerais*** [ 01/01/2018 - 01/06/2023 ]

**City:** Belo Horizonte | **Country:** Brazil

**Website:** https://ufmg.br/ | **Field(s) of study:** Computer Science: | **Final grade:** 83% | **Thesis:** High-Performance Implementation of the Irregular Wavefront Propagation Pattern on GPU using Gunrock

As a Computer Science student at the University of Minas Gerais (UFMG), I developed several projects on various low-level topics using C and C++, as well as projects related to Data Science (in Python) and Compilers (using LLVM).

My final project at my university was developed in Rust and C++, with the goal of implementing parallel and efficient algorithms that follow the IWP (Irregular Wavefront Propagation) pattern. This pattern is characterized by elements that propagate computations irregularly to their neighbors in a multidimensional space, given that a certain propagation condition is met. In the first phase, I implemented these algorithms using Rust and its great built-in parallelism features. In a second phase, the IWP was implemented with C++ and CUDA, in a GPU environment using Gunrock, a powerful CUDA library designed for creating high-performance graph

algorithms.

## WORK EXPERIENCE

***Yalo*** - United States, California

### Senior Software Engineer

[ 08/01/2024 - Current ]

I joined Yalo after working 1 year as a contractor (when I worked at Kobe, see below). Internalized by the team, I continued developing high efficiency solutions mostly related to the backend of message services, using technologies such as, Redis, Websockets and others described below. After that, I moved to the developer experience team of the company developing a wide range of tools focused on improving the overall efficiency, standardization and capabilities of other engineering teams. Working with technologies such as GitHub Actions, GitHub CI/CD, OpenAPI and so on, using mostly Typescript and Go. Now, I'm working with the core backend infrastructure of the company, creating and maintaining the core pieces of the platform. This includes message exchanging protocols, Kafka, Authentication and more.

***Kobe Apps*** - Brazil, Porto Alegre

### Software Engineer

[ 01/04/2023 - 08/01/2024 ]

I joined Kobe Apps to work at Yalo (my current company) as a contractor. After a year, I was internalized by the client and hired full-time by the Yalo team. During this 1 year period as an outsourcer, I worked on projects involving maintenance and development of backend API, networking and messaging services, using technologies such as Kafka, Redis, Websockets, Kubernetes, GCloud and so on. We developed solutions aimed at greater maintainability, efficiency and scalability of the services offered.

***Stack31*** - Brazil, Belo Horizonte

### Software Engineer

[ 01/01/2021 - 01/04/2023 ]

I've started at the company as a Golang Developer, working on a service mesh manager software. There, I worked a lot with Kubernetes, Docker and everything that involves Cloud Computing. Since the middle of 2022, I've been working on a brand new product, working as a full stack developer with Node Js, Typescript and React.

***IndWise*** - Brazil, Belo Horizonte

### Software Engineer

[ 01/06/2019 - 01/01/2021 ]

My main project at the company was a react native app in which we developed a solution for easy evaluation of item samples for industries. I have worked in it since the start of the project and was the main developer of it. We used React Native, Node js and Typescript. I've also worked on the main website, developing solutions as a full stack developer.

## LANGUAGE SKILLS

**Mother tongue(s):** Portuguese

**English**

**LISTENING:** C2 **READING:** C2 **WRITING:** C2

**SPOKEN PRODUCTION:** C1

**SPOKEN INTERACTION:** C1

**Spanish**

**LISTENING:** B1 **READING:** B1 **WRITING:** A2

**SPOKEN PRODUCTION:** A2

**SPOKEN INTERACTION:** A2

## PROGRAMMING SKILLS

**Programming Skills**

**General** (Go, TS, JS, Rust, C, C++, Python, Github)
**Games Engineering** (Computer Graphics, Low Level Engines, Unreal, Godot)
**Microservices Developer** (Docker, Kubernetes, Helm, Kafka, Redis)
**Full-Stack Web Developer** (Node, React, Mongo, HTML, CSS, Bootstrap)
**Mobile Developer** (React Native, Redux, Realm DB, Flutter)
**DevOps and DevEx** (Github Actions, CI/CD, Open API)